Computational Identification of High-Altitude Molecules Using Molecular Networking and Liquid Chromatography High-Resolution Mass Spectrometry (LC-HRMS): Developing Software by Chemists, for Chemists

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1 Abstract

Written after the results are complete. Summarize the environmental problem, research question, methods, software contribution, computational evaluation, chemical case studies, principal findings, and limitations.

2 Chapter 1: Introduction and Theoretical Background

2.1 Research Problem and Motivation

- Organic contaminants in remote, high-altitude environments
- The challenge of identifying unknown molecular features
- The need for reliable and reproducible molecular networking

2.2 Environmental and Chemical Background

- Long-range transport of organic contaminants
- High-altitude and remote surface-water environments
- Previous Wooster research and the current identification gap

2.3 LC-HRMS, Suspect Screening, and Non-Targeted Analysis

- LC-HRMS and tandem mass spectrometry
- Molecular features versus identified compounds
- Suspect screening and non-targeted analysis
- Limitations of current identification approaches

2.4 Molecular Networking and Theoretical Foundations

- MS/MS fragmentation and spectral similarity
- Cosine similarity and molecular-network construction
- Graph theory: nodes, edges, and connected components
- Network visualization and interpretation
- Limitations of spectral similarity as evidence of chemical identity

2.5 Existing Software and Research Gap

- MS-DIAL and GNPS in the existing workflow
- Documented limitations in the Wooster research workflow
- The need for software designed around chemists' needs

2.6 Research Question and Specific Aims

Central research question: What computational workflow and interface are needed to make molecular networking reliable, reproducible, and useful for Wooster's environmental HRMS research?

- Aim 1: Define the chemistry and data requirements.
- Aim 2: Develop an end-to-end molecular-networking workflow.
- Aim 3: Evaluate computational correctness, reproducibility, performance, and workflow usefulness.

- Aim 4: Apply the software to environmental HRMS case studies.

2.7 Project Scope and Thesis Contributions

- Clarify what the software will and will not do.
- Identify the expected Chemistry and Computer Science contributions.
- Explain the organization of the thesis.

3 Chapter 2: Methodology, Research Design, and Software Development

3.1 Overall Research Design and Workflow

- Overall research approach and data flow
- Relationship between the chemistry and computational components
- Research datasets and evaluation strategy

3.2 Environmental Samples and Chemistry Experimental Methods

- Sample origins, locations, and sample types
- Instrumentation and analytical conditions
- LC-HRMS and MS/MS acquisition
- Positive- and negative-ion modes, where applicable
- Clarification of work performed by previous researchers

3.3 Upstream Data Processing and Chemical Annotation

- MS-DIAL processing and version
- Feature detection, alignment, and spectral exports, as applicable

- Data-quality checks and inclusion criteria
- Suspect screening and candidate annotations
- Input data preparation and metadata
- Identification confidence and limitations

3.4 Software Requirements and System Architecture

- Functional and nonfunctional requirements
- Supported input formats and validation
- System modules and data flow
- Data structures and metadata representation
- Reproducibility and provenance

3.5 Spectral Comparison and Similarity Calculation

- Spectrum representation and preprocessing
- Peak matching and mass tolerances
- Cosine similarity implementation
- Edge thresholds and computational considerations
- Handling missing, incomplete, or unusable spectra

3.6 Molecular-Network Construction

- Node and edge representation
- Similarity thresholds and graph filtering
- Connected components and isolated features
- Linking network elements to chemical metadata

3.7 User Interface Design and Interaction

- Network layout and visual encodings
- Node and edge selection
- Feature metadata and similarity inspection
- Filtering, navigation, and network exploration
- Saving, exporting, or persistence, where implemented
- Design decisions informed by chemist workflows

3.8 Software Implementation and Reproducibility

- Programming languages, libraries, and dependencies
- Modular design, version control, and documentation
- Error handling and logging
- Configuration, parameter recording, and reproducible execution

3.9 Software Evaluation and Testing Methods

- Unit, integration, and end-to-end tests
- Reference calculations for similarity validation
- Graph-construction correctness
- Repeated-run reproducibility
- Runtime and scalability measurements
- Interface and workflow task evaluation

3.10 Chemical Case-Study Design (Testing the Software on Real Data)

- Selection of representative environmental samples
- Criteria for selecting samples, features, and network clusters
- Use of existing annotations and spectral evidence
- Approach to interpreting known and unidentified features
- Criteria for limiting chemical claims

4 Chapter 3: Results and Data Analysis

4.1 Environmental Dataset and Feature Summary

- Dataset characteristics and sample coverage
- Number of input features and usable MS/MS spectra
- Available annotations, exclusions, and data limitations

4.2 Software Implementation Results

- Completed workflow and supported functionality
- Implemented architecture and interface
- Representative input-to-network examples

4.3 Computational Correctness and Validation

- Similarity calculations versus reference values
- Input-validation and error-handling results
- Network-construction tests
- Reproducibility and persistence results

4.4 Software Performance and Workflow Evaluation

- Runtime and memory use, where measured
- Performance across tested dataset sizes
- Interface and workflow task results
- Requirements met, partially met, or unmet

4.5 Molecular-Network Structure and Characteristics

- Nodes, edges, connected components, and isolated features
- Similarity-score distributions
- Effects of thresholds and filtering, where evaluated

4.6 Chemical Case Study 1: Annotated Molecular Features

- Selected network and chemical context
- Spectral relationships and available annotations
- Interpretation and evidence strength

4.7 Chemical Case Study 2: Unidentified Molecular Features

- Selected unknown features or network clusters
- Relationships to annotated or other unidentified features
- Potential chemical relationships and unresolved questions

4.8 Summary of Principal Results

- Concise synthesis of the strongest computational and chemical findings

5 Chapter 4: Discussion

5.1 Interpretation and Integration of Results

- Answer the central research question using both result sets.
- Explain how chemistry informed software design and evaluation.
- Explain how the software supported chemical investigation.

5.2 Computational Contributions and Design Trade-Offs

- Reliability, correctness, reproducibility, and usability
- Strengths and weaknesses of algorithms and architecture
- Comparison with intended requirements and existing workflows

5.3 Chemical Significance and Comparison with Literature

- Interpretation of case-study findings
- Relationship to existing environmental chemistry research
- What networking reveals about unknown features
- Limits on claims about identity and transformation products

5.4 Study and Implementation Limitations

- Analytical and data-quality limitations
- Computational and scalability limitations
- Visualization and workflow limitations
- Evaluation constraints and limits on generalization

5.5 Implications for Future Environmental HRMS Research

- How the workflow may support future Wooster projects
- Additional data, validation, or software capabilities needed

6 Chapter 5: Conclusions and Future Work

6.1 Summary of the Research

Revisit the environmental problem, objectives, and approach.

6.2 Principal Computer Science Contributions

Summarize the implemented workflow and evaluation findings.

6.3 Principal Chemistry Contributions

Summarize the supported chemical findings and their significance.

6.4 Answer to the Central Research Question

State what the project demonstrated and what remains unresolved.

6.5 Future Work/Directions

Additional validation, environmental datasets, software improvements, and further chemical investigation.

6.6 Final Remarks

Conclude with the project's contribution to computational chemistry.

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A Supported Input Formats and Data Schema

B Additional Computational Methods and Test Cases

C Supplementary Software Evaluation Results

D Additional Molecular-Network Figures and Chemical Data

E Software Setup and Reproducibility Instructions

F AI Use Documentation, as Required by Departmental Policy